一颗刺果钩出的发明

The Burrs That Became Velcro



衣服上的刺果，怎样变成了日常搭扣？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1941 George de Mestral, a Swiss engineer, returned from hunting in the Alps with his hound, an Irish Pointer. Prickly burdock seed-heads had hitched themselves to his clothes and lay tangled in the dog's fur. Under a microscope he saw thousands of tiny barbs, each a hook grasping the small loops in cloth.

Few took his idea seriously at first. In Lyon, then a weaving centre, a cotton version proved too frail to survive abrasion. Working with a textiles firm, he turned to nylon, whose hooks were stiff enough to hold yet easy to pull apart. The trial-and-error at the looms and mills lasted about ten years.

The name Velcro merged two French words, velours, meaning velvet, and crochet, meaning hook. A US patent duly followed in 1955. In 1959 a New York fashion show sent apparel stitched with the fasteners down the runway. The public was less charmed than expected, mostly because of the aesthetic.

The underdog proved its utility in outer space. NASA used it in the Apollo programme to keep gear anchored in weightlessness, so that nothing would hover in mid-air. Apollo 11 was equipped with a sizeable amount, about 3,300 square inches. NASA did not invent the nylon clasp.

When the patent ran out in 1978, cut-price knock-offs from other countries overran the market, and a niche product went mainstream. In 2017, to stop its brand-name melting into a generic noun, the company asked people to call such mimics hook and loop fasteners. In effect, the firm was asking the public to stop using its own name for rivals' goods.

中文译文

1941年，瑞士工程师乔治·德·梅斯特拉尔带着自己的猎犬——一只爱尔兰指示犬——从阿尔卑斯山狩猎归来。带刺的牛蒡种子头钩在衣服上，也缠进狗毛里。他用显微镜观察，发现上面有成千上万根微小倒刺，每根都像钩子，扣住织物上的小毛圈。

起初，几乎没人认真对待他的想法。他在当时的纺织中心里昂试制棉制搭扣，但它太脆弱，经不住磨损。与一家纺织公司合作时，他改用尼龙：这种材料的钩子足够挺硬，既能扣牢，又容易拉开。织机旁、工厂里的反复试验持续了约十年。

Velcro这个名字合并了两个法语词：velours意为天鹅绒，crochet意为钩子。1955年，美国专利随之获批。1959年，纽约一场时装秀让缝有这类搭扣的服装登上T台。然而公众的反应不如预期热烈，主要原因是外观。

这件起初不被看好的发明，在外太空证明了用途。美国航天局在阿波罗计划中用它把装备固定住，避免装备在失重环境里漂浮。阿波罗11号配备的搭扣面积相当可观，约3300平方英寸。但这种尼龙扣件并非美国航天局发明。

1978年专利到期后，其他国家的廉价仿品涌入市场，小众产品也走向大众。2017年，为防品牌名逐渐变成整类产品的通用名词，公司要求人们把这类仿品称为“钩毛搭扣”。实际上，它是在请公众不要再用自家的名字称呼竞争对手的商品。

核心词汇

先记这22项。连着下方的原文短语一起回想。

barb n.
倒刺
thousands of tiny barbs

grasp v.
抓住；扣住
a hook grasping the small loops

loop n.
毛圈；环
the small loops in cloth

frail adj.
脆弱的
a cotton version proved too frail

abrasion n.
摩擦造成的磨损
to survive abrasion

nylon n.
尼龙
he turned to nylon

stiff adj.
挺硬的；不易弯曲的
hooks were stiff enough to hold

pull apart 动词短语
拉开；扯开
easy to pull apart

trial and error n.
反复试验；试错
The trial-and-error

merge v.
合并
merged two French words

apparel n.
服装
apparel stitched with the fasteners

fastener n.
扣件；搭扣
apparel stitched with the fasteners

aesthetic n.
外观；审美风格
mostly because of the aesthetic

utility n.
实用性；用途
proved its utility in outer space

anchor v.
固定住
keep gear anchored in weightlessness

hover v.
悬浮；漂浮
hover in mid-air

equip with 动词短语
配备
was equipped with a sizeable amount

run out 动词短语
期限届满
the patent ran out in 1978

niche adj.
小众的
a niche product went mainstream

mainstream adj.
主流的；大众化的
a niche product went mainstream

generic adj.
通用的；非特定品牌的
a generic noun

in effect 副词短语
实际上
In effect, the firm

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

Swiss adj.
瑞士的

hound n.
猎犬

prickly adj.
多刺的

seed-head n.
含种子的花序；种子头

hitch v.
钩住；挂住

tangle v.
缠结

fur n.
动物皮毛

hook n.
钩子

take v.
与seriously连用，认真对待

seriously adv.
认真地

weaving n.
织造；纺织

textile n.
纺织品

loom n.
织机

mill n.
工厂；纺织厂

velvet n.
天鹅绒

duly adv.
如期；按预期

stitch v.
缝上

runway n.
时装表演的T台

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

charm v.
使着迷；吸引

mostly adv.
主要地；大多

underdog n.
不被看好的一方

outer space 名词短语
外太空

Apollo programme 专有名词
阿波罗计划

gear n.
装备

in weightlessness 介词短语
在失重环境中

mid-air n.
半空中

sizeable adj.
相当大的；可观的

clasp n.
扣件

cut-price adj.
廉价的；降价的

knock-off n.
仿制品

overrun v.
大量涌入；占满

brand name n.
品牌名

melt into 动词短语
逐渐融为；变成

noun n.
名词

mimic n.
仿制品

rival n.
竞争对手

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Under a microscope he saw thousands of tiny _____,

02 he turned to _____, whose hooks were stiff enough to hold yet easy to pull apart.

03 The underdog proved its _____ in outer space.

04 a niche product went _____.

文章主线

- 启发 瑞士工程师梅斯特拉尔发现刺果用微钩扣住衣物毛圈，便仿照它设计可反复扣合、拉开的搭扣。
- 试制 棉制搭扣不耐磨，他改用尼龙，反复试验约十年，1955年获得美国专利。
- 应用 公众主要因外观而反应冷淡；美国航天局在阿波罗计划中用搭扣固定失重中的装备，并非搭扣的发明者。
- 普及 1978年专利到期后，廉价仿品涌入市场，原本小众的搭扣逐渐成为日常用品。
- 正名 Velcro是品牌名；2017年品牌方要求将仿品称为钩毛搭扣，避免自家商标泛指整类产品。

参考答案　1 barbs 2 nylon 3 utility 4 mainstream